Happy October 13th :)

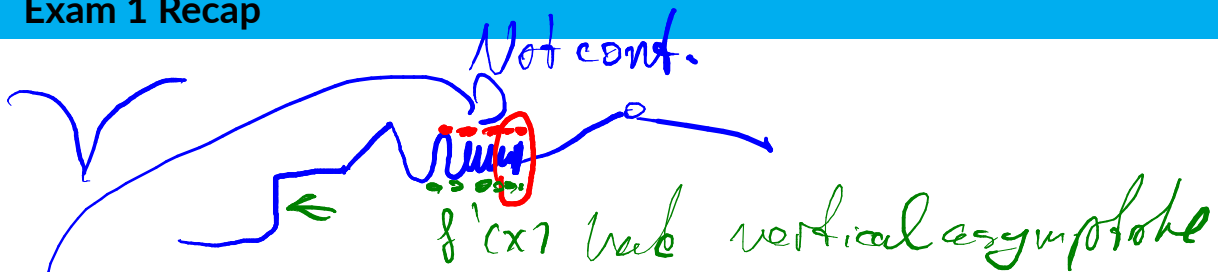
Lecture 10 - Product/Quotient Rule, Trig Derivatives

Garrett Credi

10/13/2026

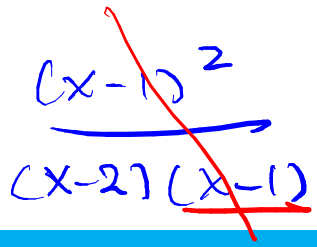
Tomorrow, make sure to wish Maxwell a happy Gotcha Day!

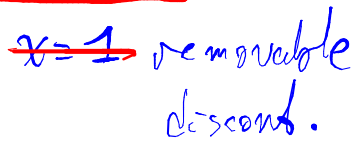
Exam 1 Recap



1. All grades for Exam 1 released, regrade requests open until next Monday.
2. After Monday, curve will be determined
3. Recap: Essential Discontinuity, Vertical Tangent Singularity, Vertical Asymptotes

Also, Anonymous Feedback Form.





Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz :)

$$\frac{d}{dx}\left(f(x) \pm g(x)\right) = \frac{d}{dx}\left(f(x)\right) \pm \frac{d}{dx}\left(g(x)\right)$$

✓

Product and Quotient Rule

Long Term Goals

Recall we're trying to build a *synthetic* approach to studying derivatives:

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Atomic Rules:

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Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

Long Term Goals

Recall we're trying to build a *synthetic* approach to studying derivatives:

Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Product Rule I

We want to determine what $\frac{d}{dx}(f(x) \cdot g(x))$ is, hopefully in terms of simpler derivatives involving $f(x), g(x)$.

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Is it just $f'(x)g'(x)$?

$\frac{d}{dx}(x^2) = 2x$ ✗

$\neq$

$\frac{d}{dx}(x) \cdot \frac{d}{dx}(x) = 1 \cdot 1 = 1$

Product Rule I

We want to determine what $\frac{d}{dx}(f(x) \cdot g(x))$ is, hopefully in terms of simpler derivatives involving $f(x), g(x)$.

Is it just $f'(x)g'(x)$? NO NO NO NO NO!

$\frac{d}{dx}(f(x)g(x))$

$\frac{d}{dx}(x^2) = 2x$

$2x \quad 1+1$

$1 \quad 1 \quad 1$

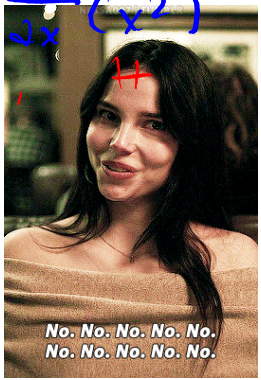


Figure: My one wish willow is for you to not compute derivates of products like this!

Product Rule II

Well, then what is it?

Product Rule II

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Approach 1: Linearization!

functions derivative linear thing

Product Rule II

Well, then what is it?

Approach 1: Linearization! We motivated the other rules by examining what the operation did to linear functions first - let's try that here!

Product Rule II

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Linearization of $f(x)$ at $x = x_0$: $f(x_0) + f'(x_0)(x - x_0)$,

Linearization of $g(x)$ at $x = x_0$: $g(x_0) + g'(x_0)(x - x_0)$,

Product Rule II

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Linearization of $f(x)g(x)$ at $x = x_0$:

$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \quad (1)$$

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Linearization of $f(x)g(x)$ at $x = x_0$:

$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \quad (1)$$

$$= f(x_0)g(x_0) + f'(x_0)g(x_0)(x - x_0) + f(x_0)g'(x_0)(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \quad (2)$$

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$$= f(x_0)g(x_0) + (f'(x_0)g(x_0) + f(x_0)g'(x_0))(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \quad (3)$$

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So why is it NOT $f'(x)g'(x)$?

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So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!

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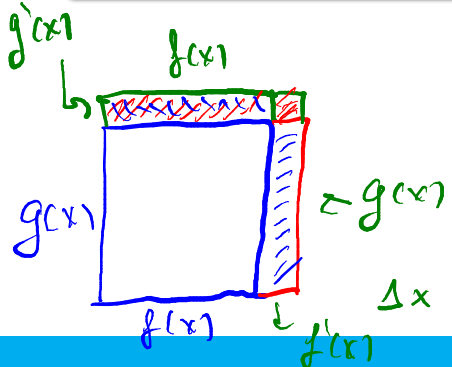
So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!

The derivative would be the slope on the *linear* term.

Product Rule III

The Product Rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x) \tag{4}$$



$f(x)g'(x) +$

$f'(x)g(x)$

Product Rule Examples I

Example 1: What is the derivative of:

$$x^2 e^x + \frac{2^x}{x}? \tag{5}$$

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$$= (\frac{d}{dx}x^2) \cdot (e^x) + (x^2) \cdot (\frac{d}{dx}e^x) + \frac{d}{dx}(2^x \cdot \frac{1}{x}) \tag{7}$$

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$$= (2x) \cdot e^x + x^2 \cdot e^x + (\frac{d}{dx}2^x) \cdot (\frac{1}{x}) + (2^x) \cdot (\frac{d}{dx}\frac{1}{x}) \tag{8}$$

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$$= 2xe^x + x^2e^x + \ln(2)2^x \cdot \frac{1}{x} + 2^x \cdot \frac{-1}{x^2} \tag{9}$$

Product Rule Examples II

For a more applied example:

Product Rule Examples II

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Product Rule Examples II

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Suppose we've measured the relevant quantities at:

$$m(1) = 50mi, \quad e(1) = \frac{1}{25}\frac{gal}{mi}, \quad p(1) = 4.15\frac{\$}{gal} \tag{10}$$

$$m'(1) = 0\frac{mi}{day}, \quad e'(1) = \frac{1}{1000}\frac{\frac{gal}{mi}}{day}, \quad p'(1) = 0.01\frac{\frac{\$}{gal}}{day} \tag{11}$$

Product Rule Examples II

Words => Math

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no quotient

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What is the rate of change of the cost to drive your car on day 1?

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Product Rule Examples III

Step 1: Cost Function.

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$$= m'(t)e(t)p(t) + m(t)e'(t)p(t) + m(t)e(t)p'(t) \tag{14}$$

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Product Rule Examples IV

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$$= 0 \cdot \frac{1}{25}\frac{gal}{mi} \cdot 4.15\frac{\$}{gal} \tag{16}$$

$$+ 50mi \cdot \frac{1}{1000}\frac{\frac{gal}{mi}}{day} \cdot 4.15\frac{\$}{gal} \tag{17}$$

$$+ 50mi \cdot \frac{1}{25}\frac{gal}{mi} \cdot 0.01\frac{\frac{\$}{gal}}{day} \tag{18}$$

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$$= 0\frac{\$}{day} + 0.2075\frac{\$}{day} + 0.02\frac{\$}{day} \tag{19}$$

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$$= 0.2275\frac{\$}{day} \tag{20}$$

Quotient Rule I

Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

max list

max time

Quotient Rule I

Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

Is it just $\frac{f'(x)}{g'(x)}$?

max 1 rule

max time

$\frac{x^3}{x^2} = x \rightarrow 1$

$\frac{3x^2}{2x} = \frac{3}{2}x$

Quotient Rule I

Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

Is it just $\frac{f'(x)}{g'(x)}$? SUPER NO! NO! NO!

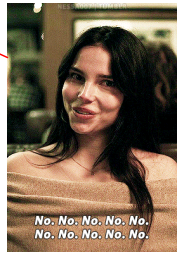


Figure: I'm amending my one wish willow to also include you not computing derivatives of quotients like this.

Quotient Rule II

We can determine what it actually is by realizing that:

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{21}$$

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$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{21}$$

and since we know how to linearize $f(x)$ and how to linearize a product, we need only understand how to linearize something like $\frac{1}{g(x)}$.

First, we note that for the derivative of $\frac{f(x)}{g(x)}$ at $x = x_0$ to exist, the function $\frac{f(x)}{g(x)}$ must be continuous at $x = x_0$, so $g(x_0) \neq 0$.

This allows us to approximate $\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)}$.

Quotient Rule III

Since

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)}$$

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$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)} = \frac{1}{(g(x_0) - g'(x_0)x_0) + g'(x_0)x} \tag{22}$$

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we can focus on understanding how to linearize functions like

$$\frac{1}{a + bx}. \tag{23}$$

We compute:

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Quotient Rule IV

And so,

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0) \tag{27}$$

Quotient Rule IV

And so,

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Quotient Rule V

Moral of the story: it's HARD to actually get the derivative of a quotient

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Quotient Rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{32}$$

Why $g(x)^2$?
1) have $1/g(x)$
2) decreasing my f' & g' by $g(x)$

$\frac{-g'(x)}{g(x)^2}$

$\frac{\text{low} \cdot d\text{High} - \text{high} \cdot d\text{low}}{\text{low}^2}$

Quotient Rule Example I

E.g. computing:

$$\frac{d}{dx}\frac{2xe^x + 3x^2}{x^3 + 2x + 1} \tag{33}$$

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$$= \frac{(\frac{d}{dx}(2xe^x) + \frac{d}{dx}(3x^2)) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{35}$$

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$$= \frac{(2e^x + 2xe^x + 6x) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{36}$$

Trigonometric Derivatives

Derivative of $\sin(x)$ I

Recall from our discussion on the Squeeze Theorem:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \tag{37}$$

$$\lim_{x \to 0} \frac{\cos(x) - 1}{x} = 0 \tag{38}$$

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What is this limit in terms of derivatives?

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$\rightarrow \frac{d}{dx}\Big|_{x=0} \cos(x)$

What is this limit in terms of derivatives?

$$\frac{d}{dx}\Big|_{x=0} \sin(x) = 1, \frac{d}{dx}\Big|_{x=0} \cos(x) = 0 \tag{39}$$

Derivative of $\sin(x)$ II

What about the derivative function?

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$$\frac{d}{dx}\sin(x) = \lim_{h\to 0}\frac{\sin(x+h) - sin(x)}{h} \tag{40}$$

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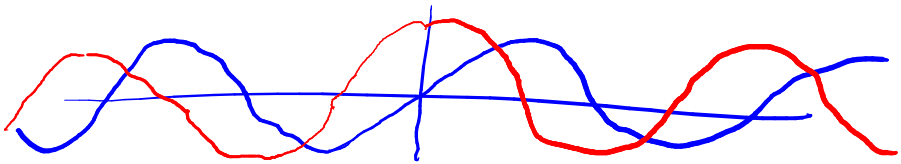
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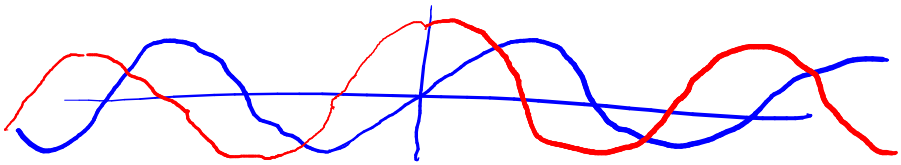
Derivative of $\sin(x)$ III



Derivative of $\sin(x)$

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{46}$$

Derivative of $\sin(x)$ III



Derivative of $\sin(x)$

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{46}$$

What about for $\cos(x)$?

Derivative of $\cos(x)$ I

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Derivative of $\cos(x)$

Derivative of $\cos(x)$ I

$$\frac{d}{dx}\cos(x) = \lim_{h\to 0}\frac{\cos(x+h)-\cos(x)}{h} \tag{47}$$

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0

1

Derivative of $\cos(x)$

Derivative of $\cos(x)$ I

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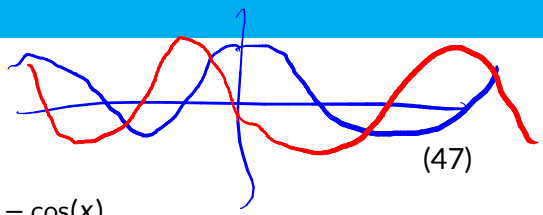
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0 1

Derivative of $\cos(x)$

$$\frac{d}{dx}\cos(x) = -\sin(x) \tag{50}$$

Derivative of $\tan(x)$

What about for $\tan(x)$? Do we need to do even more limit yuckiness?

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No! Since $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$ we can just use quotient rule!

$$\frac{d}{dx}\tan(x) = \frac{d}{dx}\frac{\sin(x)}{\cos(x)} \tag{51}$$

$$= \frac{\left(\frac{d}{dx}\sin(x)\right)\cdot\cos(x) - \sin(x)\cdot\left(\frac{d}{dx}\cos(x)\right)}{\cos(x)^2} \tag{52}$$

$$= \frac{\cos(x)\cdot\cos(x) - \sin(x)\cdot(-\sin(x))}{\cos(x)^2} \tag{53}$$

$$= \frac{\cos(x)^2 + \sin(x)^2}{\cos(x)^2} \tag{54}$$

Derivative of $\tan(x)$

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$$= \frac{1}{\cos(x)^2} = \sec(x)^2 = 1 + \tan(x)^2 \tag{55}$$

Derivative of $\sin(2x)$

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$$= 2 \cdot (\cos(x)^2 - \sin(x)^2) \tag{60}$$
$$= 2 \cdot \cos(2x) \tag{61}$$

Worksheet Time!

Let's get into groups (or if you really like working alone, please stay so I can be here to help :)) and work on the worksheet!

End of Class!

That’s all folks!